

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 39)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain why a moderate water shortage reduces crop productivity even though the plants survive.

[3 marks]

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0 1 . 2

Explain the problems caused by waterlogged soils and describe **two** ways farmers reduce soil water levels.

[4 marks]

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0 1 . 3

Explain how irrigating with groundwater can cause soil salinisation and how this harms crops.

[2 marks]

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0 1 . 4

State how a farmer could raise the pH of an acidic soil and explain why a low pH reduces fertility.

[1 mark]

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10

0 2

Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha ⁻¹ yr ⁻¹	Food energy output / GJ ha ⁻¹ yr ⁻¹
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 2 . 1

Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

0 2 . 2

Use your answers to explain the difference between productivity and efficiency.

[3 marks]

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0 2 . 3

Explain what is meant by an **energy subsidy** and give **three** examples.

[3 marks]

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0 2 . 4

Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

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10

0 3 . 1

Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

[3 marks]

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0 3 . 2

Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

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03 . 3

Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

03 . 4

Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

1

2

10

04 . 1

Explain how living organisms provide each of the following ecosystem services.

[6 marks]

Regulation of atmospheric composition

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Soil maintenance

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Interspecies relationships

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04 . 2 Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk. **[2 marks]**

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04 . 3 Explain why elephants are particularly important seed dispersers. **[2 marks]**

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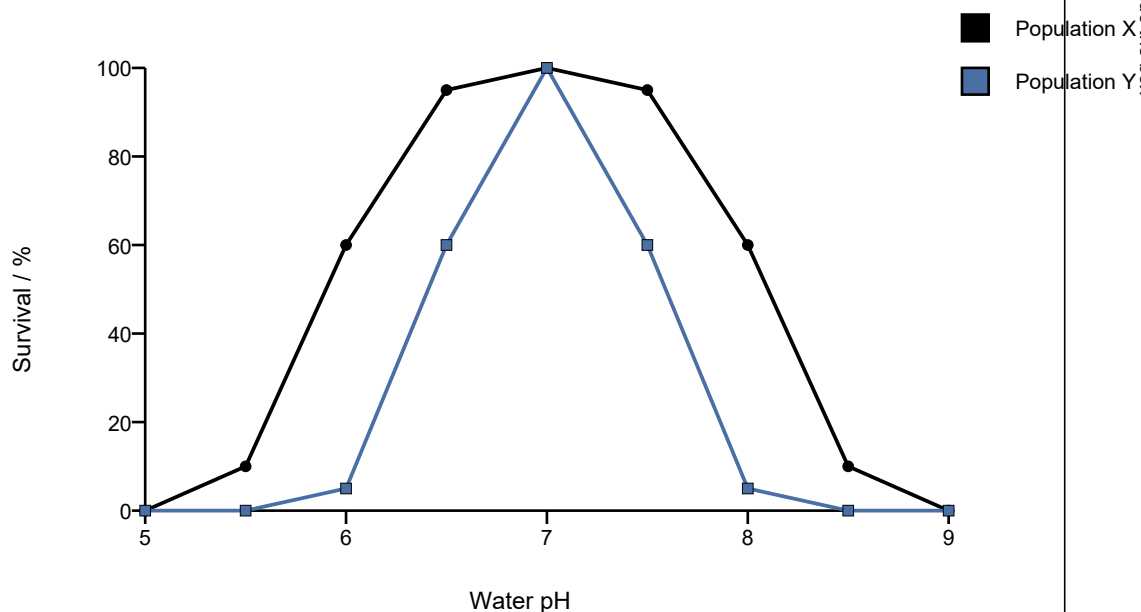
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10

0 5

Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

Figure 1



0 5

1

Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

[3 marks]

0 5

2

A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

[3 marks]

0 5 . 3

Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

0 5 . 4

Give **two** biotic factors that affect the distribution of a species.

[2 marks]

1

2

10

0 6 . 1

Describe the ecological features of temperate broadleaf woodland and explain how woodland floor plants such as bluebells are adapted to them.

[4 marks]

0 6 . 2

Explain why an ancient woodland (existing before 1600) usually has higher biodiversity than a recently planted woodland.

[3 marks]

06 . 3

Describe **three** threats to UK broadleaf woodland.

[3 marks]

1

2

3

10

07 . 1

Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

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07 . 2

Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

1

2

3

0 7 . 3 Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Isotope analysis of ice cores

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Gas analysis of ice cores

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Radiocarbon dating

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Magnesium : calcium ratio in calcite

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10

0 8 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station

☐

B Flooding a valley to create a reservoir

☐

C Lowering the water table by groundwater abstraction

☐

D Acid mine drainage entering a stream

☐

0 8 . 2 Give **four** human activities that cause complete habitat destruction.

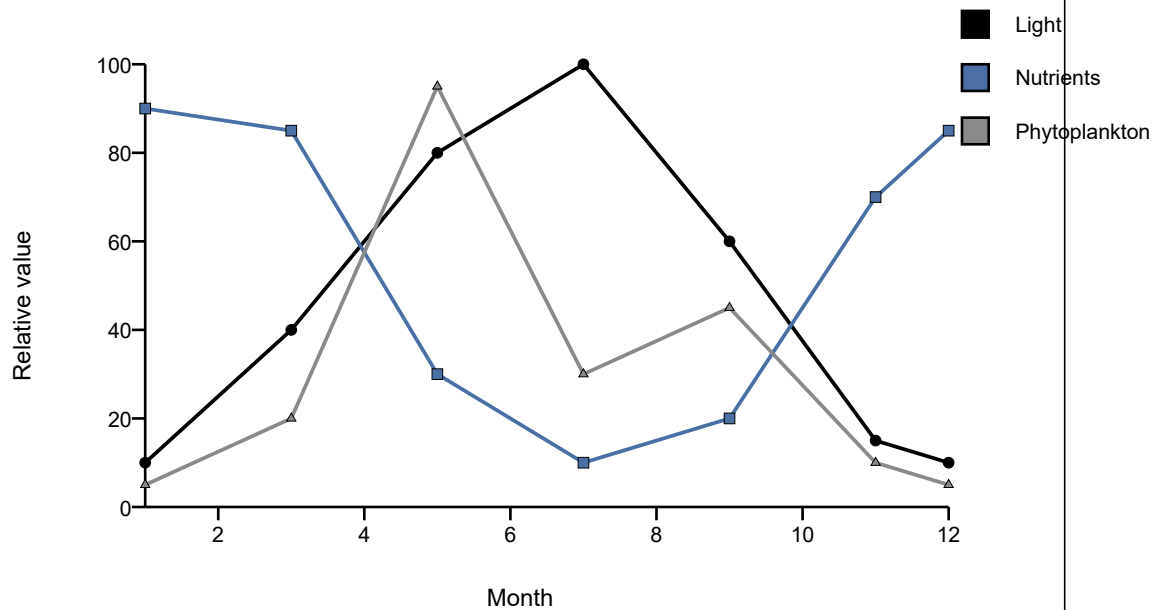
[4 marks]

- 1
- 2
- 3
- 4

5

0 9 Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



0 9 . 1 Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

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09 . 2

Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]

09 . 3

Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]

09 . 4

Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]

10

1 0 . 1

Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

Do not write outside the box

Extra space

10 . **2** State what is meant by a **fed** aquaculture species.

[1 mark]

10

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

1 1 . 2

Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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[illegible]

END OF QUESTIONS